

AISSAOUI Ismail

AI & Data Science Engineer

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PROFILE

AI & Data Science Engineer specialized in Edge AI, deep sequence models (Mamba, xLSTM), and performance optimization (C++ JIT, 0.04ms latency). Proven track record in designing hybrid architectures and Physics-Informed Digital Twins (PINN) for mission-critical industrial systems. Passionate about solving complex engineering problems at the intersection of hardware constraints and algorithmic design.

SKILLS

Programming: Python, TypeScript, JavaScript, Java, SQL

ML / DL: PyTorch, TensorFlow, Keras, Scikit-learn, Lightning, ONNX

AI Techniques: Transformers (BERT, GPT, RAG), GNN, GAN, XGBoost, LSTM

Optimization: PSO, Genetic Algorithm, ACO, Simulated Annealing, Bayesian Optimization

Web & API: React, Next.js, FastAPI, Node.js, TailwindCSS

Data & MLOps: MLflow, Airflow, Kubeflow, DVC, Pandas, NumPy

Databases: PostgreSQL, MongoDB, MySQL, Neo4j, Cassandra

Cloud & DevOps: Docker, Linux, Git, CI/CD, CUDA, AWS, Grafana

Soft Skills: Technical leadership, Communication, Stakeholder management

PROFESSIONAL EXPERIENCE

Graduate AI Research Engineer (Intern)

Jan 2026 – Jul 2026

Sonatrach (Industrial Surveillance Division) — Algeria

Led the architectural design and deployment of an AI-native Digital Twin platform for monitoring critical industrial assets (Gas Turbines).

- Developed a Physics-Informed Digital Twin using state-of-the-art **Mamba-SSM** and **xLSTM** architectures.
- Solved the "Latency-vs-Precision" trade-off through hardware-software co-design, achieving **0.04ms inference time** on edge devices via C++ JIT optimization.
- Implemented custom PINN loss functions to ensure thermodynamic consistency in generative surrogates.
- Designed quantitative reliability indicators for Out-of-Distribution (OOD) anomaly detection.

Network Engineering Intern

Sep 2024

Algérie Télécom RMC Center — Chlef, Algeria

EDUCATION

National Higher School of Computer Science (ESI-SBA)

2021–2026

State Engineer's Degree and Master's Degree in Artificial Intelligence

CERTIFICATIONS

DeepLearning.AI: Machine Learning Specialization, Deep Learning Specialization, NLP Specialization

ZTM Academy: PyTorch: Zero to Mastery, TensorFlow Developer

PROJECTS

Physics-Informed Digital Twin (Mamba-SSM/xLSTM)	2026
Generative surrogate architecture achieving 0.04ms edge inference for industrial gas turbines. <i>Tech: PyTorch, C++, CUDA — Concepts: Mamba-SSM, PINN, Edge Latency Optimization</i>	
GNN-based Recruiting System (Career Connect)	2024–2025
CV/job matching using graph embeddings and relational learning. <i>Tech: PyTorch Geometric, SBERT, FastAPI, PostgreSQL — Concepts: GNN, Similarity, Embeddings</i>	
Aura — Enterprise AI Trading System	2025
Full-stack automated crypto trading platform powered by hybrid LSTM forecasting and Reinforcement Learning (Deep Q-Learning) to optimize entry/exit policies under long-term reward maximization. Includes advanced technical indicators, real-time dashboards, and interactive backtesting. <i>Tech: TensorFlow, Keras, FastAPI, Next.js, WebSockets — Concepts: LSTM, Reinforcement Learning, Deep Q-Learning, Time-Series Analysis</i>	
Advanced NLP Pipeline with Multi-Optimization	2025
Multilingual moderation optimized with PSO, Genetic Algorithm, and Bayesian Optimization. <i>Tech: DistilBERT, PyTorch, FastAPI — Concepts: Optimization, Transformers, Classification</i>	
Intelligent University Scheduler	2025
Automatic generation of conflict-free timetables using hybrid heuristics and evolutionary algorithms. <i>Tech: FastAPI, NumPy, Pandas — Concepts: GA, PSO, ACO, Simulated Annealing</i>	
Mini-GPT (Decoder-only)	2025
Implementation of a compact GPT model with controlled training and sequential text generation. <i>Tech: PyTorch, CUDA, AdamW, W&B — Concepts: Transformers, Language Modeling</i>	
North African Sentiment BERT	2025
LoRA adaptation of BERT for sentiment analysis in Maghrebi dialect. <i>Tech: PyTorch, LoRA, MLflow — Concepts: Fine-tuning, Low-Rank Adaptation</i>	
Geo-RAG — Map Assistant	2025
RAG system integrating geospatial data for map-oriented queries. <i>Tech: ChromaDB, GeoPandas, Streamlit — Concepts: RAG, Spatial Reasoning</i>	
Recognizini — Facial Recognition	2025
Facial recognition with incremental embeddings and an active feedback loop. <i>Tech: OpenCV, PyTorch, Redis, FastAPI — Concepts: Embeddings, Drift Management</i>	
Real-time PPE Detection (YOLOv8)	2025
Real-time detection of safety equipment (helmet, vest) for compliance monitoring. <i>Tech: YOLOv8, OpenCV — Concepts: Object Detection, Streaming</i>	
Traffic Analytics Suite	2025
Traffic analytics including tracking, speed estimation, and license plate recognition. <i>Tech: YOLOv8, ByteTrack, EasyOCR — Concepts: Tracking, OCR, Analytics</i>	

LANGUAGES

Arabic: Mother Tongue | English: Advanced | French: Conversational